

Prediction of Air Quality Index (AQI) Using Regression Models

Abstract

Objective of the Study:

The goal of this report is to predict a continuous target variable, Air Quality Index (AQI), using regression techniques.

Data Description:

Air quality data forms the core here - it tracks daily levels **like PM2.5, PM10, CO, NOx, O3, and SO2**. Numbers pile up across several columns, each feeding into calculations for AQI. Linked to global aims through the UNSDG framework, especially Goal 3 about healthier lives. Breathing conditions shape well-being; that connection drives this work forward.

Purposed Methodology:

The methodology includes Exploratory Data Analysis (EDA), building regression models including a Neural Network, hyperparameter optimization, feature selection, and final model comparison.

Key Findings:

One after another, the models got tested with MAE, MSE, RMSE, and R^2 score. Turns out, methods built by grouping models together did better than straight-line predictions when guessing air quality levels.

Overall Summary:

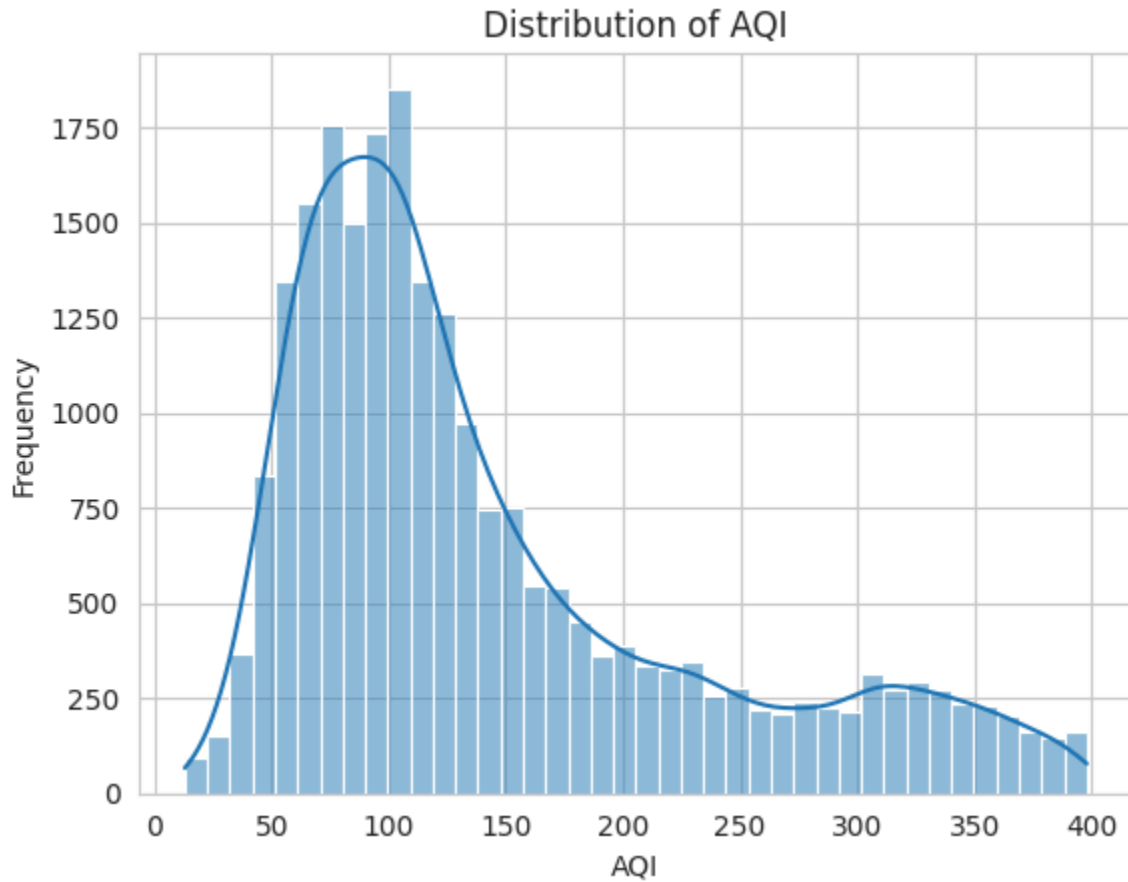
The final models achieved strong predictive performance, with Random Forest Regression providing the best accuracy. Key insights include the importance of feature selection and hyperparameter tuning in improving model generalization.

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Figure List

- AQI distribution plot



Shows the spread and frequency of Air Quality Index (AQI) values in the dataset, helping to identify skewness, central tendency, and the presence of extreme pollution levels.

- Model performance comparison table

Table 2: Comparison of Final Regression Models

Model	Features Used	Test RMSE	Test R ²
Ridge Regression	Selected (6)	35.86	0.842
Random Forest Regressor	Selected (6)	27.42	0.907

This table presents a comparative evaluation of the final regression models using key metrics such as RMSE and R² score on the test dataset. It highlights the relative effectiveness of each model in predicting AQI and supports the selection of the best-performing model.

1. Study Foundation

1.1 Problem Statement & Context

The goal of this project is to predict a continuous target variable, Air Quality Index (AQI), using pollutant concentration data. Accurate AQI prediction helps assess environmental conditions and supports public health decision-making.

1.2 Dataset Characteristics

The dataset used in this analysis is an Air Quality dataset obtained from an open environmental data source. It contains numerical pollutant

measurements such as PM_{2.5}, PM₁₀, CO, NO_x, dataset aligns with **UNSDG 3 (Good Health and Well-being)** by supporting air pollution monitoring and health risk assessment.

1.3 Primary Research Goals

The objective of this analysis is to build predictive regression models that accurately estimate AQI using the given air pollution features.

2. Research Design and Workflow

2.1 Pre-processing & Data Refinement

Missing pieces got pulled out of the data, along with fixes to keep feature formats uniform. Because numbers filled every column, switching categories into codes wasn't needed at all. Scaling adjustments popped up only when certain models - like linear ones or neural networks - entered the picture.

2.2 In-depth Data Exploration (EDA)

Looking at the data through histograms, box plots, scatter plots, or correlation matrices showed clear patterns. What stood out was how closely AQI tied to pollutants like PM_{2.5} along with PM₁₀, suggesting these play a major role in shaping air quality.

2.3 Architecture & Model Construction

Task 1 – Deep Learning Approach (Neural Network)

- A feedforward neural network with multiple hidden layers was used

- ReLU activation function and Mean Squared Error
- Adam optimizer with tuned learning rate
- Train–validation split to monitor overfitting

Task 2 – Comparative Classical Algorithms

Two regression models were considered:

- **Ridge Regression**
- **Random Forest Regressor**

Both models were trained using an 80–20 train–test split.

2.4 Metrics for Success Assessment

The following metrics were used:

For Neural Network:

- MAE
 - MSE
 - RMSE
-
- R^2 Score

For Classical ML Models:

- MAE
- MSE
- RMSE
- R^2 Score

2.5 Fine-tuning & Parameter Search

Hyperparameter optimization was performed using **GridSearchCV** and **RandomizedSearchCV**.

- Ridge Regression: Optimal alpha value was selected
- Random Forest Regressor: Optimal number of estimators, depth, and split parameters were identified

2.6 Optimal Feature Identification

Feature selection was conducted using a **Filter-based approach (correlation analysis)**.

The selected features were:

PM2.5, PM10, CO, NO_x, O₃, SO₂

These features were chosen due to their strong correlation with AQI and relevance to air pollution science.

3. Analysis of Outcomes

3.1 Core Observation Summary

The models were evaluated on the test dataset using regression metrics. Random Forest Regression achieved lower RMSE and higher R^2 compared to Ridge Regression, indicating better predictive performance.

3.2 Candidate Model Selection

Based on evaluation metrics, **Random Forest Regressor** was selected as the final model, achieving:

- Test RMSE ≈ 27.4
- Test $R^2 \approx 0.91$

3.3 Deployment & Execution Hurdles

Challenges encountered include feature multicollinearity among pollutants and increased computational time during hyperparameter tuning.

3.4 Potential Enhancements

Future improvements could involve:

- Advanced ensemble models (e.g., Gradient Boosting, XGBoost)
- Time-series modeling using temporal data

- Inclusion of meteorological features

4. Synthesis of Results

4.1 Assessing Model Efficacy

Random Forest outperformed linear models due to its ability to capture non-linear relationships between pollutants and AQI.

4.2 Impact of Optimization Strategies

Hyperparameter tuning significantly improved model stability, while feature selection reduced noise and improved interpretability.

4.3 Meaning and Implications of Results

Pollutants such as PM_{2.5} and PM₁₀ were found to be the strongest predictors of AQI, aligning with real-world environmental studies.

4.4 Constraints of the Current Study

Limitations include dataset size, lack of temporal modeling, and absence of meteorological variables.

4.5 Strategic Directions for Future Inquiry

Future research may explore deep learning models, real-time AQI prediction systems, and cross-city generalization.

5. References

- World Health Organization (WHO) Air Quality Guidelines
- United Nations Sustainable Development Goals (SDG 3)